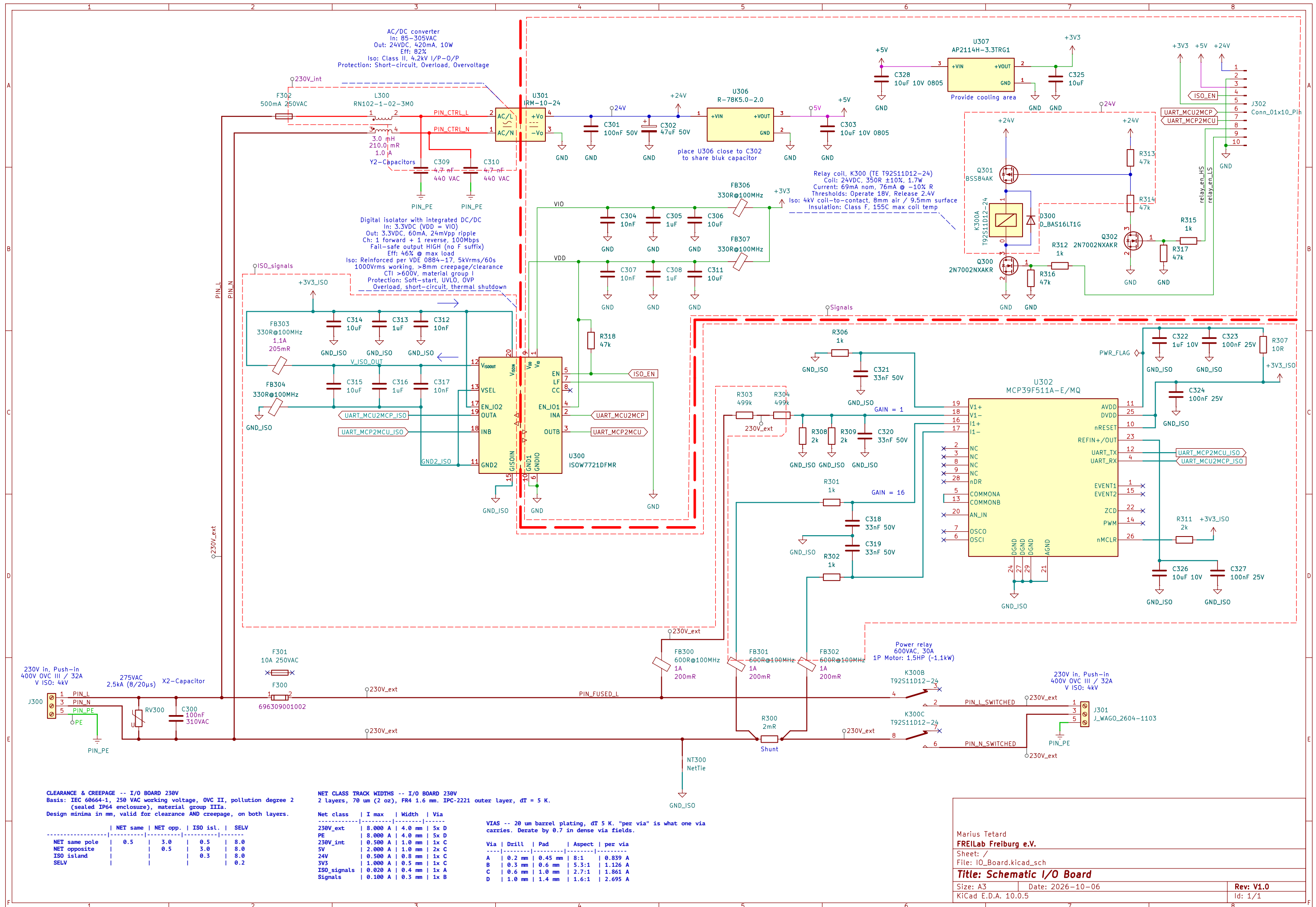




CLEARANCE & CREEPAGE -- I/O BOARD 230V  
Basis: IEC 60664-1, 250 VAC working voltage, OVC II, pollution degree 2 (sealed IP64 enclosure), material group IIIA.  
Design minima in mm, valid for clearance AND creepage, on both layers.

|               | NET same | NET opp. | ISO isl. | SELV |
|---------------|----------|----------|----------|------|
| NET same pole | 0.5      | 3.0      | 0.5      | 8.0  |
| NET opposite  |          | 0.5      | 3.0      | 8.0  |
| ISO island    |          |          | 0.3      | 8.0  |
| SELV          |          |          |          | 0.2  |

NET CLASS TRACK WIDTHS -- I/O BOARD 230V  
2 layers, 70 um (2 oz), FR4 1.6 mm. IPC-2221 outer layer, dT = 5 K.

| Net class   | I max   | Width  | Via  |
|-------------|---------|--------|------|
| 230V_ext    | 8.000 A | 4.0 mm | 5x D |
| PE          | 8.000 A | 4.0 mm | 5x D |
| 230V_int    | 0.500 A | 1.0 mm | 1x C |
| 5V          | 2.000 A | 1.0 mm | 2x C |
| 24V         | 0.500 A | 0.8 mm | 1x C |
| 3V3         | 1.000 A | 0.5 mm | 1x C |
| ISO_signals | 0.020 A | 0.4 mm | 1x A |
| Signals     | 0.100 A | 0.3 mm | 1x B |

VIAS -- 20 um barrel plating, dT 5 K. "per via" is what one via carries. Derate by 0.7 in dense via fields.

| Via | Drill  | Pad     | Aspect | per via |
|-----|--------|---------|--------|---------|
| A   | 0.2 mm | 0.45 mm | 8:1    | 0.839 A |
| B   | 0.3 mm | 0.6 mm  | 5.3:1  | 1.126 A |
| C   | 0.6 mm | 1.0 mm  | 2.7:1  | 1.861 A |
| D   | 1.0 mm | 1.4 mm  | 1.6:1  | 2.695 A |

Marius Tetard  
**FREILab Freiburg e.V.**  
Sheet: /  
File: IO\_Board.kicad\_sch  
**Title: Schematic I/O Board**  
Size: A3 Date: 2026-10-06 Rev: V1.0  
KiCad E.D.A. 10.0.5 Id: 1/1